

Triple-drug Fixed-dose Combination of Dapagliflozin + Glimepiride + Metformin Extended Release: Rationale for Combination

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Epidemiology of Diabetes in India

- Globally, India ranks 2nd in the total number of diabetes cases¹
- Diabetes prevalence was 74.2 million in 2021 and is expected to increase to 124.9 million by 2045^{1,2}

Unmet needs in Management of Type 2 Diabetes Mellitus

- Chronic high glucose levels can cause glucose toxicity in several organs leading to complications such as chronic kidney disease, heart disease, nerve damage^{3,4}
- With diabetes, there is a progressive decline in beta-cell function. Therefore, many patients will eventually require multiple antihyperglycemic agents⁵
- Dual therapy (Metformin + SU/SGLT2i + Metformin) with mean baseline HbA1c of 8%–8.9% has shown < 22% patients are able to achieve target HbA1c < 7%
- In addition, high pill burden and complex treatment regimens decrease adherence⁶

Need for Fixed Dose Combinations (FDC)

Diabetes has multifactorial pathophysiologies⁷.
Therefore, components of FDC should:

Act by complimentary mechanisms⁸

Should be well tolerated as FDC⁸



Triple-drug FDCs effectively achieve glycemic targets⁷

Guideline Recommendations for Inadequately Controlled Diabetes with Metformin

2023 ADA Recommendations⁹

- Pharmacologic approaches that provide the efficacy to achieve treatment goals should be considered, such as metformin or other agents, including combination therapy, that provide adequate efficacy to achieve and maintain treatment goals

2020 RSSDI-ESI Clinical Practice Recommendations¹⁰

- If glucose targets are not achieved with two agents: start **third oral agent**-AGI, DPP-4 inhibitor, SGLT2 inhibitor, or TZDs (depending on second-line agent used) or start insulin or GLP-1 agonists

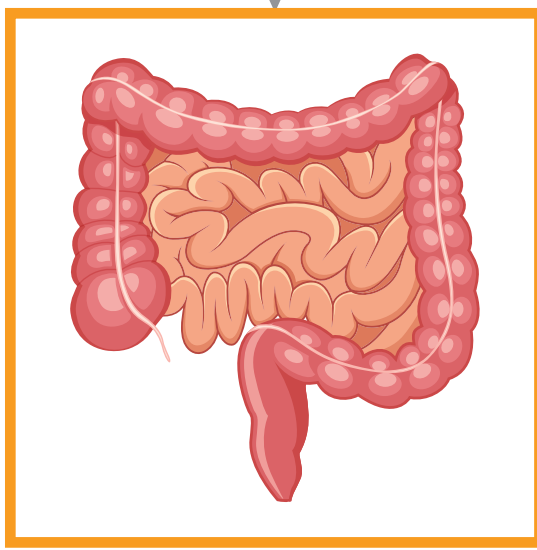
Drug Profile – Metformin, Glimepiride, and Dapagliflozin

Metformin – First-line Treatment for > 50 Years¹¹

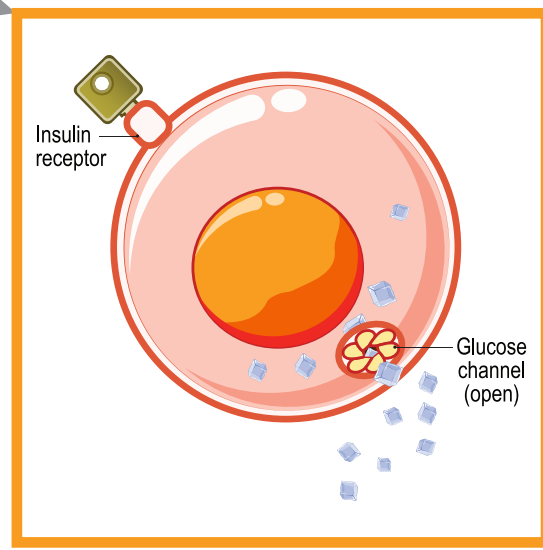
Mechanism



↓ Glucose production¹¹



↓ Glucose absorption¹²



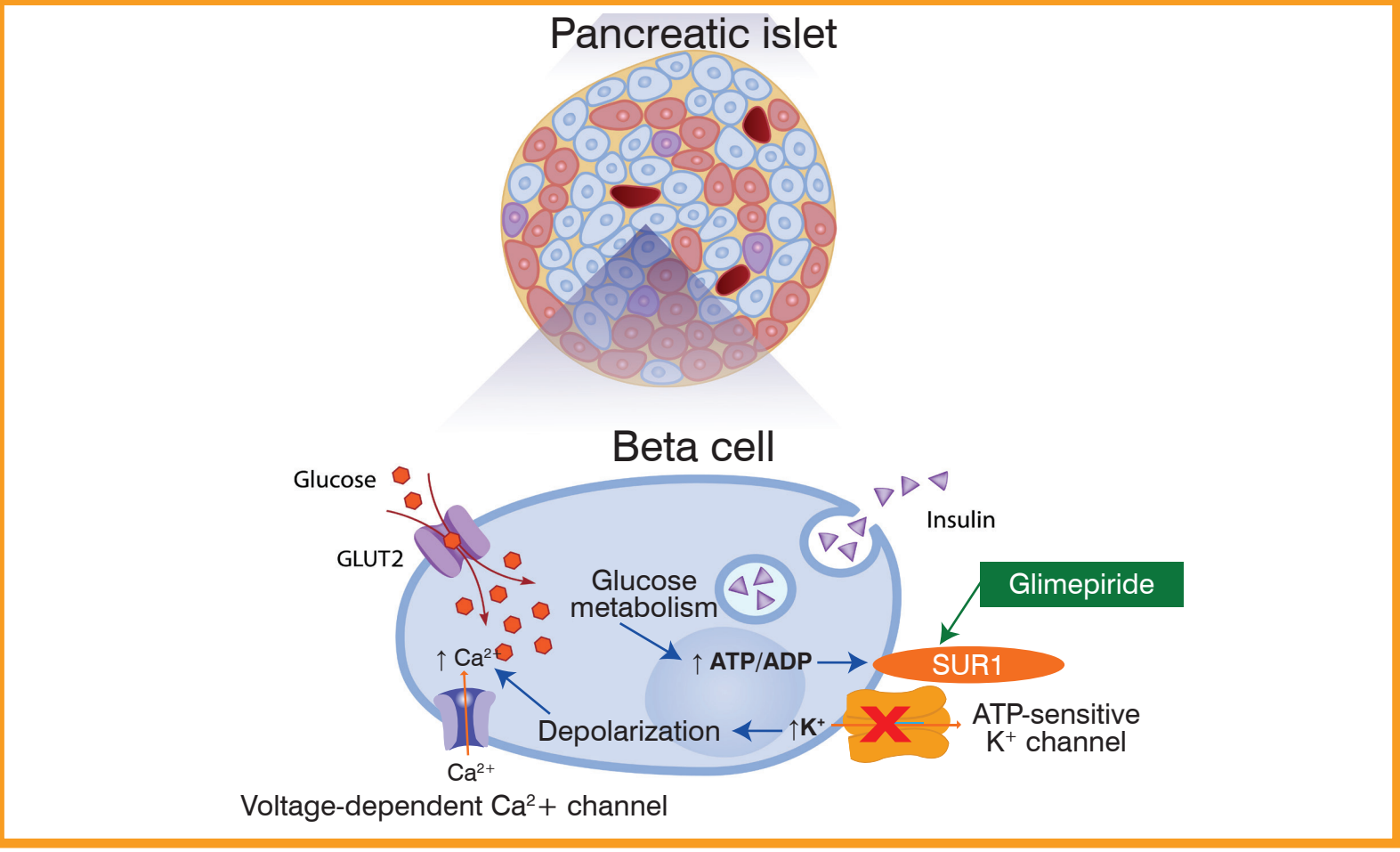
↑ Insulin sensitivity¹³

- Improves insulin resistance¹¹
- Demonstrated weight loss/weight neutrality¹⁴
- CV benefits¹⁴

- Reduces HbA1c without increasing hypoglycemia risk¹⁴
- 1.5%–2.2% ↓ in HbA1c¹⁵

Glimepiride – A Modern SU¹⁶

Increases insulin secretion through stimulation of pancreatic beta cells^{17,18}



Compared to other SUs¹⁷

- Lesser weight gain
- Lower risk of hypoglycemia

Proven CV safety¹⁹

Rapid lowering of FPG²⁰

1.5%–2.0% ↓ in HbA1c¹⁷

Dapagliflozin – A Highly Selective SGLT2i²¹

- Insulin-independent mechanism²¹
- Improves glycemic control through glycosuria with associated caloric loss leading to a reduction in body weight²¹
- Minimal risk of hypoglycemia²¹
- ↓ CV death or hospitalization due to HF with or without prior ASCVD²²
- ↓ CKD progression²³

0.5–0.7% ↓ in HbA1c²⁴

Triple-drug FDC

Advantages of triple-drug FDC

- Allows once-daily dosing²⁵
- Drugs with complimentary mechanism of action²⁵
- Enhance weight loss in overweight/obese patients⁷
- Cardiovascular benefits⁷

Dapagliflozin + Glimepiride + Metformin FDC

- ↑ insulin sensitivity¹³
- ↓ insulin resistance¹¹
- ↓ body weight²¹
- 21% more patients achieved target HbA1c

Evidence for Dapagliflozin + Glimepiride + Metformin (Data from 24-week Randomized, Double-blind, Placebo-controlled, Parallel-group, International, Multicenter Studies)

Phase 3 study²⁶

- **Dapagliflozin vs Placebo**
HbA1c reduction from baseline(p < 0.0001):
 - Dapagliflozin (2 mg) = **-0.58%**
 - Dapagliflozin (5 mg) = **-0.63%**
 - Dapagliflozin (10 mg) = **-0.82%**
- Sustained **reduction in body weight** from baseline:
 - Dapagliflozin (2 mg) = -1.18 kg (p < 0.1410)
 - Dapagliflozin (5 mg) = -1.56 kg (p < 0.0091)
 - Dapagliflozin (10 mg) = -2.26 kg (p < 0.0001)

Phase 3b study²⁷

- **Dapagliflozin (10 mg/day) vs Placebo**
 - **Reduction in HbA1c** from baseline (**-0.86%** vs -0.17%, p < 0.0001)
 - **Improvement in FPG** from baseline (-34.2 mg/dl vs -0.8 mg/dl, p < 0.0001)
 - **Reduction in body weight** from baseline (-2.7 kg vs -0.6 kg, p < 0.0001)

Both the studies showed that Dapagliflozin when used in a combination therapy was effective, well-tolerated, and reduced body weight.^{26,27}

Conclusion

Triple-drug FDC (Dapagliflozin + Glimepiride + Metformin) will provide better safety and efficacy due to:

- Complimentary mechanisms of action
- Improved compliance
- Additional benefit of weight loss
- Potential for reducing complications
- Less risk of hypoglycemia

FDC: Fixed dose combination; SGLT2i: Sodium/glucose cotransporter-2 inhibitors; SU: Sulfonylurea; AGI: Alpha-glucosidase inhibitors; DPP4: Dipeptidyl peptidase 4 inhibitors; GLP-1: Glucagon-like peptide-1; TZD: Thiazolidinedione; American Diabetes Association; RSSDI-ESI: Research Society for the Study of Diabetes in India; HbA1c: Glycated hemoglobin; CV: Cardiovascular; SUR1: Sulfonylurea receptor 1; ATP: Adenosine triphosphate; ADP: Adenosine di phosphate; GLUT2: Glucose transporter; Calcium ion; K+: Potassium ion; HF: Heart failure; ASCVD: Atherosclerotic cardiovascular disease; FPG: Fasting plasma glucose; CKD: Chronic kidney disease

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